

A. Medically Relevant Findings from the Article

Stage-specific molecular signatures:

The study revealed that systemic lupus erythematosus (SLE) has distinct transcriptomic profiles during the establishment phase and the exacerbation phase, suggesting that different molecular mechanisms dominate at different stages.

Cell-type-specific contributions:

Certain immune cells (e.g., dendritic cells, B cells) play a central role early in disease onset. Other cells (e.g., monocytes/macrophages, T cells) become more active during disease flare-ups.

Key role of interferon and inflammatory pathways:

Interferon signaling and inflammation-related genes showed distinct expression dynamics across stages, explaining immune dysregulation and flare mechanisms in SLE.

Clinical significance:

Identified molecular markers that can distinguish disease phases, which may help guide personalized treatment.

Suggested potential therapeutic targets, such as stage-specific immune pathways that could be modulated to slow disease progression.

B. Genomic Techniques Used

RNA sequencing (RNA-seq):

Used to profile gene expression in different immune cells from patient blood samples.

Cell-type-resolved transcriptomics:

Immune cell subsets (e.g., T cells, B cells, monocytes) were isolated and analyzed separately to capture cell-specific signals.

Bioinformatics approaches:

Differential gene expression (DGE) analysis.

Gene co-expression network analysis.

Pathway enrichment analysis to identify dysregulated signaling pathways.

C. New Research Questions / Hypotheses

Longitudinal dynamics of transcriptomic changes

Question: Do the transcriptomic signatures identified here dynamically shift within the same patient across remission and flare?

Rationale: If so, these markers could serve as predictive biomarkers to anticipate flares and enable preemptive treatment.

Functional validation of key genes and pathways

Question: Are the stage-specific genes and pathways causal drivers of disease progression, or merely downstream byproducts?

Rationale: Experimental validation in animal models or via CRISPR-based perturbation would clarify therapeutic potential.

Predicting treatment response

Question: Can transcriptomic profiles at establishment vs exacerbation predict response to current therapies (e.g., corticosteroids, B-cell depletion, interferon inhibitors)?

Rationale: This could enable precision medicine by tailoring therapy to disease stage and patient-specific immune profiles.

Intercellular communication networks (optional expansion)

Question: How do different immune cells interact across stages (e.g., B cells and monocytes exchanging signals)?

Rationale: Mapping communication networks may reveal synergistic intervention points beyond single-gene targeting.